

Jackie Chan

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Education

University of Illinois Urbana-Champaign

Ph.D. in Computer Science; Advisor: Dr. Eshwar Chandrasekharan

Thesis Title: Quantifying the Impacts of Algorithmic Ranking on Social Media Trending Feeds

Urbana, IL

Sept. '20 – May '26

Carleton College

B.A. in Computer Science & Mathematics; Cum Laude

Northfield, MN

Sept. '16 – June '20

Experience

Machine Learning Research Associate (Intern), Wealth Management

New York, NY

Morgan Stanley

June '26 – Aug. '26

- Developed and evaluated graph-based machine learning approaches for matching financial advisors to financial products, leveraging explicit relationships among advisors, clients, and accounts to serve as a challenger to an internal gradient-boosted tree model.
- Constructed a heterogeneous graph with 3M+ nodes and 7M+ edges from multiple internal data sources and developed, trained, and tuned an attention-based graph neural network using PyTorch Geometric (PyG); achieved comparable ROC-AUC and lift to the production model.

Graduate Research Assistant

Urbana, IL

University of Illinois Urbana-Champaign

Sept. '21 – May '26

- Led computational social science research at the Social Computing Laboratory, designing and executing large-scale studies on algorithmic ranking and online governance using machine learning, causal inference, and natural language processing.
- Recognized with a National Science Foundation (NSF) GRFP Honorable Mention (2022) for excellence in research potential and proposal quality.
- Taught CS courses in programming and data visualization across 12 semesters; received Outstanding TA Award (2021).

Publications

The Ranking Effect: How Algorithmic Rank Influences Attention on Social Media

ACM Conference on Human Factors in Computing Systems (CHI '27, Under Review)

- Led a large-scale randomized user experiment ($n = 585$) to quantify the causal impact of algorithmic ranking on user engagement and content perception.
- Engineered a custom full-stack web application (React, FastAPI, MongoDB) with automated CI/CD (GitHub Actions, Vercel) to facilitate real-time data collection and maintenance.

<https://arxiv.org/abs/2509.18440>

Examining Algorithmic Curation on Social Media: An Empirical Audit of Reddit's r/popular Feed

Proceedings of the International AAAI Conference on Web and Social Media (ICWSM '26)

- Built and deployed a high-availability scraper (Python, Docker) that captured 220K+ snapshots and 130K+ distinct posts over 11 months, managing a multi-terabyte data pipeline.
- Integrated and applied a BERT-based transformer model to perform sentiment and toxicity analysis on 5M+ Reddit comments, quantifying antisocial behavior patterns across the platform.
- Developed a causal inference framework using Bayesian inference and stratified propensity score matching to model the bidirectional relationship between antisocial behavior and feed positioning.

<https://doi.org/10.1609/icwsm.v20i1.42644>

Understanding Community Resilience: Quantifying the Effects of Sudden Popularity via Algorithmic Curation

Proceedings of the International AAAI Conference on Web and Social Media (ICWSM '24)

- Quantified the impact of sudden popularity on the moderation health of 1,320 Reddit communities, analyzing surges in comment volume, removal rates, and newcomer participation.
- Applied longitudinal time-series analysis to identify factors that contributed to community resilience following large-scale attention spikes.

<https://doi.org/10.1609/icwsm.v18i1.31310>

Technical Skills

Programming Languages: Python, SQL, TypeScript, JavaScript, Java, Kotlin, HTML/CSS

Frameworks & Technologies: PyTorch, PyTorch Geometric, pandas, NumPy, PySpark, scikit-learn, PyMC, React, FastAPI, PostgreSQL, MongoDB, Docker, Git, Linux, Vercel